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**Topic:** Predictors of Entry Into Medication-Assisted Therapy Based on Diagnoses of Opioid Dependence

**Objective:** To determine which factors predict entry into treatment for opioid dependence using medication-assisted therapy (i.e. use of opioid replacement medication such as methadone or buprenorphine). Exploratory data analysis and initial logistic regression models will be built to evaluate key metrics. A final logistic regression model will then be created to determine the predominant factors.

**Data Source, Explanatory PDF, and Details:**

<https://www.samhsa.gov/data/data-we-collect/teds-treatment-episode-data-set/datafiles>

[https://www.samhsa.gov/data/system/files/media-puf-file/TEDS-A-2022-DS0001-info-codebook\\_v1.pdf](https://www.samhsa.gov/data/system/files/media-puf-file/TEDS-A-2022-DS0001-info-codebook_v1.pdf)

|                       |           |                         |          |
|-----------------------|-----------|-------------------------|----------|
| Full dataset records: | 1,498,034 | Full dataset size:      | 708.6 MB |
| DSM dataset records:  | 296,907   | DSM dataset size:       | 68.0 MB  |
| Post-cleaned records: | 162,352   | Post-cleaned data size: | 29.6 MB  |

Due to the sheer number of attributes in this dataset, only a select few key attributes will be listed below. A full breakdown of the entire datasets variables can be found in the PDF linked above directly under the *Data Source, Explanatory PDF, and Details* subtitle. Note that numerous variables were re-classified/categorized to reduce the number of dummy variables entering the final logistic regression model. These re-classified/categorized variables will not be included under engineered attributes/features.

**Key Attributes/Features:**

- METHUSE: Medication-assisted opioid therapy at admission (dependent variable)
- DSMCRIT: DSM Diagnosis (used to define opioid dependence study population)
- GENDER: Gender
- AGE: Age at admission
- RACE: Race
- ETHNIC: Ethnicity
- LIVARAG: Living arrangement
- PSOURCE: Referral source
- DIVISION: Census division
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**Engineered Attributes/Features:**

- NON\_OPIOID\_DRUGS\_FLG: All non-opioid drug flags combined. If a single non-opioid drug was used, this column would reflect that.
- OPIOIDS\_FLG: All opioid drug flags combined in a similar manner to non-opioid drugs flag

The dataset is of particular interest to the team given the interest and work in the field of epidemiology and pharmaceuticals by a team member, and the current opioid crisis present

across North America. The dataset provides a chance to apply data science and statistical techniques learned in the course, while also learning more about this subject area.

### **Methods:**

The full dataset was initially explored using Python's Pandas and Numpy modules before pairing it down to only those with a DSM diagnosis of opioid dependency or abuse (opioid dataset). Comparative analysis was then performed between the full dataset and the opioid dataset, including gender proportions and gender proportions with respect to wait times. Missing data became a large obstacle as the dataset was quite cleansed between variables/attributes, making imputation a challenge. To reduce the number of attributes in the DSM dataset, two attributes were engineered, an opioids flag and non-opioid drugs flag, respectively.

After looking into the types of missingness, several attempts at various imputation strategies were performed, including imputation by mode and simply using *missing* as a category. Multivariate imputation methods were considered, but require skillsets above what was learned in the course. For several variables, scikit-learn's SimpleImputer function was used to do mode imputation.

Data was reclassified where possible to reduce the number of categories per column, which in turn reduced the amount of missing data. Once reclassification/categorization was complete, two logistic regression models were created to determine if dropping the missing data would substantially harm the results. This was not the case, so the missing data was dropped from the DSM dataset.

A third logistical regression model was run utilizing subject knowledge, literature, and the results of the exploratory analysis to determine key variables to add. The model was built using an additive approach. The odds and confidence intervals of each variable in the model were calculated, and interpretations of the results were made.

### **Analysis:**

**Challenge 1:** During the data exploration phase heatmaps were produced using Python's seaborn module. The missing data heatmap showed large swaths of missing data present in blocks of rows throughout a large number of variables/attributes. Upon further investigation into the missingness types, health insurance appeared to be missing at random, while those corresponding to the blocks of rows appeared to be missing systematically by institutions. Imputation and reclassification of variables was performed to mitigate as much of the missing data as possible, but methods beyond the scope of those taught in this class would need to be learned and used to try and resolve more of the missing data.

**Challenge 2:** While it may have been easier to do a subtractive model, the sheer number of variables, especially nominal variables that were originally broken down into dummy variables, was too much for statsmodel to handle. Scikit-learn's LogisticRegression and classification\_report functions were then tried, but the output report did not include p-values, nor a guiding pseudo-R<sup>2</sup>. For this reason statsmodel was used with an additive approach,

however, this required having prior knowledge of the subject matter to build an informed model

**Conclusions:**

The analysis found some significant factors associated with medication-assisted treatment for people diagnosed with an opioid use dependency. The predominant factors include division and health insurance, while race and gender were less influential. Data quality was very high facilitating the cleaning process, however, the large volume of what appears to be systematic missing data proved challenging to overcome, limiting the overall scope and analysis potential of the dataset. The logistic regression modeling using exclusively categorical data required adapting approaches learned in the course with online materials since the course mainly focused on the use of continuous variables and traditional linear regression.